Plate, is subducted (i.e., thrust) beneath the smaller Mariana Plate that lies to the west.

53. The long gravity tsunami waves are caused by two interacting processes.

There is the slope of the sea surface which creates a horizontal pressure force. Then there is the piling up or lowering of sea surface as water moves in varying speeds in the direction that the wave form is moving. These processes together create propagating waves. As a tsunami leaves deep waters and propagates into the shallow waters, it transforms. This is because as the depth of the water decreases, the speed of the tsunami reduces. But the change of total energy of the tsunami remains constant. Tsunamis behave as shal- low-water waves because of their long wavelengths. They have a period in the range of ten minutes to two hours and a wavelength exceeding 500 km.

54.

- When a region is dissected by streams into a system of valleys and divides, submergence produces a highly irregular shoreline called ria coastline. The coast of south-west Ireland is a typical example of ria coastline.
- Some coastal regions have been heavily eroded by glacial action and the valley glacier troughs have been excavated below sea level. After the glaciers have disappeared, a fjord coastline emerges. These coasts have long and narrow inlets with very steep sides. The fjord coasts of Norway are a typical example.

55. Laccolith formed by intrusive volcanism and Mushroom Rocks formed due to Aeolian Erosion are most common mushroom shaped rocks formed by geomorphic processes. Though mushroom rocks can be formed by glacial erosion, such rocks formed by glacial deposition are not common. Marine deposition usually does not lead to formation of mushroom rocks.

56. In arid regions occasional rainstorms produce numerous rills and channels which extensively erode weak sedimentary formations. Ravines and gullies are developed by linear fluvial erosion leading to the formation of badland topography. Example: Chambal Ravines. Powerful wind continuously abrades stone lattices, creating holes. Sometimes the holes are gradually widened to